Report outline — at a glance

IAA AI-Assisted Hackathon 2026 · Woojin Ji & Krista Stanke · draft Oct 20, for Thursday 7 pm · 5 pages, covers Part 1 and Part 2

AI writes the code; actuaries own the assumptions.

AI made coding and breadth of analysis fast. It did not make the assumptions right — transparent models, rules fixed in advance and human questions did.

Three points that support it

1 Transparent by default

GLM with readable rating tables. A complex model only where it passed rules set in advance.

Evidence: GBM +0.009 D^2 in buildings, but safest 10% over-priced by 15–30% → kept GLM.

2 AI for code, people for assumptions

Code is cheap to check (tests). Assumptions are not (questions + experiments).

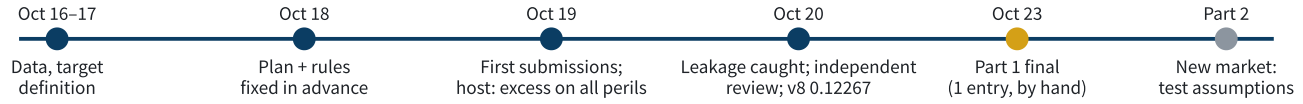
Evidence: a stronger reviewer model's two largest claims were rebutted by our data.

3 Part 2 tests the assumptions

Every assumption has a "would be wrong if" condition, checked on the new market.

Evidence: the assumption ledger (page 2), right-hand column filled in Part 2.

Timeline



Structure: process order, Part 1 → Part 2 inside each section

Section	Part 1 (now)	Part 2 (later)
0 Summary Krista	Thesis, 3 points, scores	*Add Part 2 score*
1 Data & target Woojin	Issues found & fixed (one table)	*Checks rerun; new issues*
2 Assumptions Woojin	How each was checked; BI reform; ledger	*Ledger: held / broke / re-estimated*
3 Model Woojin	GLM; measured GBM trade-off	*What changed for the new market*
4 Validation both	Rules; 7 of 9 predictions; leakage	*Same rules reused?*
5 AI use Krista	Roles; counts from AI log; errors caught	*Part 2 entries*
6 Limits & lessons both	BI trend (+5.1% if adopted), IBNER, tail range	*Lessons across two markets*

Proposed owners — to agree on Thursday.

Page budget (5 pages)



The biggest block is the assumptions — the centre of the thesis. Judging weights: Pricing 40% · AI 25% · Trust 20% · Communication 15%.

The spine: assumption ledger (section 2)

One row per key assumption. Part 1 columns are filled now; the last column is filled in Part 2.

Assumption	Why we believe it	Part 1 evidence	Would be wrong if	Part 2
Excess on every peril	Contract term, per claim	Host answer	—	?
Late claims via completeness	Reporting delay differs by peril	Two estimators within 0.3 pts	Reporting speed changes	?
2023 frequency step (0.93 / 0.95)	A change in 2023, then flat	Seen inside every group	Year effect vanishes in a GLM	?
Trends on gross loss, excess fixed	Inflation + leverage	Net rises faster than gross	A one-off step	?
BI reform 0.83, no later trend	Legal change from a loss date	Step 0.78–0.80; closed claims +1.2%	Step by report date only	?
Cap + Pareto loading	Heavy tail is chance, not rating	α stable; range 1.2–3.8	Tail not Pareto	?
Weather = location climate	Next year's weather unknown	Leaderboard +0.0026	Year factor changes nothing	?
Claim history predicts	Experience rating	Under-priced 22–80% before	Gradient disappears	?
Service contacts = leakage	Caused by claims	12–72× same-year	—	?

Two figures

Figure 1 — BI reform step
Mean severity by loss quarter, all vs closed claims. Shows a real cost step in 2023 Q1, and that the later "rise" is open reserves.

Figure 2 — GLM vs GBM, buildings
A/E by decile. Shows what the extra accuracy costs: the safest customers over-priced.

AI section in numbers

Count AI_USE_LOG entries by type:
AI proposed · Team questioned · AI error
Split AI errors into **code** (caught by tests and checks) vs **assumptions** (caught by questions and data). This turns the thesis into data.

How the conclusion is worded

Avoid
"AI did less than we expected." · "We followed what AI told us at first." · "GBM is also predictable."

Say
"Effective use of AI meant knowing where to trust it. Code is cheap to verify; assumptions are not." · "AI proposed the plan; the team set the rules and decided."

To decide on Thursday (7 pm)

#	Decision	Proposal
1	Thesis and three points	As above
2	Owners and draft deadline	As in the structure table; drafts before Part 2 opens
3	Figures	The two above, at most one more
4	Video (5 min)	Krista records: thesis → ledger → model trade-off → AI roles